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On the Growth of Wind-Generated Waves in a Swell-Dominated Region in the South Atlantic

The growth of surface gravity waves is investigated in an open sea region dominated by swell. An extensive database was collected by the Brazilian Oil Company during the South Atlantic Deep Water Program (PROCAP) in Campos Basin, off coast of Rio de Janeiro. This is the most important petrolic basin in Brazil where tens of drilling platforms are located. The data set contains over 5800 directional wave spectra measured by a heave-pitch-roll buoy in conjunction with meteorological data. A spectral approach is applied in a novel method for the partitioning of sea from swell and for the adjustment of the spectral parameters. The wave growth is investigated and the regression laws of the windsea extracted from the swell-contaminated spectra are compared to earlier works in sheltered areas. It is shown that, on the grounds of the equilibrium range theory, the high-frequency spectral level (the alpha parameter) scale with the reciprocal wave age in the same manner as in unimodal swell-free spectra. [DOI: 10.1115/1.1423636]

Introduction

Detailed knowledge of the shape of the ocean wave spectrum and its growth is important information for offshore engineering purposes. Among numerous examples are the estimation of induced loads on marine structures and the response of floating bodies to the wave action. In the last 50 years, several works have sought to properly describe and parameterize the wave spectrum. Most of them have produced spectral models that aim to describe only the windsea part of the spectrum, generated by the local wind, hence, they are unimodal. Furthermore, the greatest part of these studies is based on measurements on wave tanks in laboratories or in swell-free environments with well-defined fetches like lakes or sheltered coastal areas. Undoubtedly, such work has clarified the understanding of the physical mechanisms behind wave generation and growth, but there is a lack of studies in open sea regions. From the operational perspective it is important to confirm whether the theoretical relationships, for example correlating wave energy and wind, apply to the most common cases, i.e., areas in the open ocean where swell is present. The modification of the windsea growth by swell is not completely clear and the results from laboratory waves show some disagreements among experiments. It is clear, however, that there is a reduction in the amplitude of laboratory wind waves when longer (swell) waves are added.

In the North Atlantic, Dobson et al. [1] found no significant influence on the fetch laws proposed by Donelan et al. [2] after removing the swell from the spectra. However, both works were carried out in short-fetched coastal areas, and some questions still remain about the influence of swell on windsea growth in the open ocean.

It is fundamental to be able to properly separate swell from windsea in order to study whether or how the interactions between them modify the growth relations of wind waves obtained in swell-free environments. Spectral methods are necessary for the partitioning and adjustment of the multi-modal wave spectra in order to determine how energy distributes in frequency and direction. Most of the known spectral formulations are for the adjustment of unimodal spectra, and so it is not possible to apply them

for the common cases when more than one spectral peak is present. To overcome this limitation, a technique for the representation and partitioning of multi-peaked (up to three peaks) spectra is proposed.

Spectral Formulations

The spectrum measured in a particular point of the ocean is the sum of wave systems generated by events separated either in space, in time, or both. Although the wave spectrum is quite complex, there is a remarkable similarity of its shape in different locations under an enormous variety of wind speeds, spanning from small waves in lakes up to storm waves generated by hurricanes. It is believed that the evolution of the waves in deep water is a function of three dynamical processes [3]: the nonlinear wave-wave interaction (S_{nl}), the gain of energy from the wind input (S_{in}), and the dissipation due to wave breaking (S_{ds}). Among the three processes, the nonlinear interactions play a very important role in the shape stabilization of the wave spectrum $S(f)$, forcing its high-frequency portion beyond the peak frequency to decay in a manner inversely proportional to frequency to a power n in the form $S(f) \propto f^{-n}$.

The similarity hypothesis proposed by Kitaigorodskii [4] enabled the description of the development of the wave spectrum under fetch-limited conditions. A key argument in his hypothesis was the existence of a high-frequency "equilibrium range." In this equilibrium range, the dynamical processes balance each other; i.e., $S_{nl} + S_{in} + S_{ds} = 0$, rates of energy gain due to wind input are balanced by energy loss due to breaking and nonlinear wave-wave interactions, and hence the spectrum remains stable. Phillips [5] suggested that in the portion of higher energy within the equilibrium range (in the band of frequencies from $1.5f_p$ to $3.0f_p$ where f_p is the peak frequency), the spectrum is a function of f^{-5} decay expressed as

$$S(f) = \alpha g^2 (2\pi)^{-4} f^{-5} \quad (1)$$

where f is frequency, g is the gravitational acceleration, and α is a constant scale parameter set equal to 0.0081. Phillips' work was based on dimensional considerations and on the fact that due to the wave steepness, the high-frequency part of the spectrum is limited by wave breaking. Any wind input in these frequencies is lost in wave breaking, and hence (1) describes the high-frequency part of the wave spectrum.

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Pierson and Moskowitz [6] proposed a spectral form for fully developed seas from analysis of measurements in the North Atlantic. They added an exponential term to Phillips' expression (1) to generate a low-frequency spectral face

$$S(f) = \alpha g^2 (2\pi)^{-4} f^{-5} \exp\left[-\frac{5}{4}\left(\frac{f}{f_p}\right)^{-4}\right]. \quad (2)$$

The Pierson-Moskowitz spectrum (2) describes the waves developed under no limitations in fetch and time; but, fully developed seas are not very common in nature, as they require a steady wind blowing for a long period over a large area. Seeking to clarify the development of growing seas under fetch-limited conditions, researchers from Europe and USA carried out the Joint North Sea Wave Project (JONSWAP). The location of these extensive observations was carefully chosen based on the simple situations suggested by Kitaigorodskii, i.e., a steady offshore wind over a strait beach in a region not contaminated by swell. Hasselmann et al. [7] proposed the JONSWAP spectrum that described fetch-limited waves very well using five parameters (f_p , α_j , γ_j , σ_a , and σ_b)

$$S(f) = \alpha_j g^2 (2\pi)^{-4} f^{-5} \exp\left[-\frac{5}{4}\left(\frac{f}{f_p}\right)^{-4}\right] \gamma_j^{\exp[-(f-f_p)^2/2\sigma^2 f_p^2]} \quad (3)$$

where

$$\sigma = \begin{cases} \sigma_a & \text{for } f \leq f_p \\ \sigma_b & \text{for } f > f_p. \end{cases}$$

The JONSWAP spectrum (3) is based on the Pierson-Moskowitz spectrum with an enhancement factor γ_j added. This enhancement is only significant in the region near the spectral peak; and at higher frequencies, the decay is inversely proportional to frequency to the fifth power, as suggested by Phillips [5]. The peak frequency f_p is the frequency at the maximum of the spectrum. The left and right-side widths of the peak region are represented, respectively, by σ_a and σ_b . The high-frequency scale parameter α_j is not constant, as initially proposed by Phillips [5]. Hasselmann et al. [7] found a relationship between α_j and fetch which confirmed the suggestion of Longuet-Higgins [8] that α_j decreases with increasing fetch X , given by

$$\alpha_j = 0.076 \left(\frac{Xg}{U_{10}^2} \right)^{-0.22}$$

where U_{10} is the wind speed at 10 m height. They found no correlation of γ_j , σ_a , and σ_b with fetch and proposed mean values for their representation ($\gamma_j = 3.3$, $\sigma_a = 0.07$, and $\sigma_b = 0.09$).

The high-frequency form $S(f) \propto f^{-5}$ present in the JONSWAP spectrum has been questioned by later researches (see, for example, Phillips [9] that reanalyzed the equilibrium spectral balances coming up with a f^{-4} power law rather than f^{-5}). Toba [10] postulated a different formulation for the high-frequency tail represented by a f^{-4} decay which seems to be more consistent with observations. Based on detailed field measurements in Lake Ontario, Donelan et al. [2] proposed a new spectral form

$$S(f) = \alpha_d g^2 (2\pi)^{-4} f_p^{-1} f^{-4} \exp\left[\left(\frac{f}{f_p}\right)^{-4}\right] \gamma_d^{\exp[-(f-f_p)^2/2\sigma^2 f_p^2]}. \quad (4)$$

The Donelan spectrum (4) is a modified version of the JONSWAP spectrum to allow a f^{-4} frequency tail. Unlike Hasselmann et al. [7] who presented spectral parameters dependent on fetch, Donelan et al. [2] parameterized them in terms of the reciprocal wave age U_c/c_p , where U_c is the component of the wind in the wave direction and c_p is the phase at the peak frequency. For the high-frequency scale parameter, they proposed

$$\alpha_d = 0.006 (U_c/c_p)^{0.55}, \quad 0.83 < U_c/c_p < 5.$$

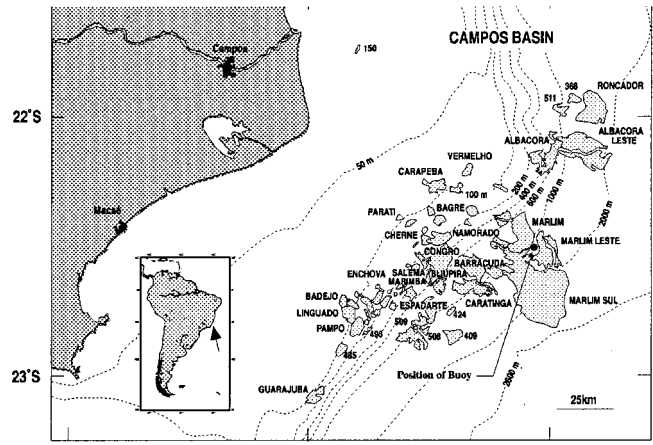


Fig. 1 Position of Campos Basin in the coast off Rio de Janeiro, Brazil, and the wave buoy. The shaded areas are the petroleum fields.

Donelan et al. [2] suggested that, in contrast to Hasselmann et al. [7], γ_d and σ depend on wave age

$$\gamma_d = \begin{cases} 1.7, & 0.83 < U_c/c_p < 1; \\ 1.7 + 6.0 \log(U_c/c_p), & 1 \leq U_c/c_p < 5 \end{cases}$$

and

$$\sigma = 0.08 [1 + 4/(U_c/c_p)^3]; \quad 0.83 < U_c/c_p < 5.$$

The exponent n of the high-frequency tail that best describes the spectrum decay $S(f) \propto f^{-n}$ is still an open question. In many studies [2,10–14], the value of n varied between -3.5 and -5 . Rodriguez et al. [15], following Kitaigorodskii [16] considerations, point out the existence of a transitional frequency at which the decay changes from f^{-4} to a f^{-5} tail. These works raise the question of whether there is a universal high-frequency decay in the form $S(f) \propto f^{-n}$, but also give strong evidence that n lies between -4 and -5 .

Data Analysis

Campos Basin, off Rio de Janeiro (Fig. 1), is the most important petrolic basin in Brazil. More than 70 percent of the petroleum prospected in the country comes from the tens of platforms located in this area. Every day, several offshore operations take place in the region, which operatively depend on the sea state, in particular, on the surface waves. This is one of the most important regions in Brazil, with huge urban concentrations and strong commercial, industrial, and touristic activities.

The South Atlantic Deep Water Program (PROCAP) was an extensive experiment carried out by the Brazilian Oil Company PETROBRAS in Campos Basin to study the main oceanographic features of the region. In terms of wind and wave measurements, a heave-pitch-roll buoy was moored in a depth over 1000 m at position $22^\circ 31' S$ and $39^\circ 58' W$ from March 1991 to March 1993 and from January 1994 to July 1995. The wind speed and direction were measured hourly from the buoy by two Young propeller-vane anemometers at a height of 3.78 m and 4.43 m, and later converted for the standard height of 10 m. Each measurement is the average of a 10-min record, obtained one per hour.

The wave buoy acquired three time series, the vertical displacement (η) and two slopes (η_x and η_y) in the east and north directions, respectively. These time series are the starting point for the spectral analysis performed. The displacements were sampled at a rate of 1 Hz during 20 min eight times a day every 3 h. The spectral estimators were obtained using the Welch method (see, for example, Marple Jr. [17]). Records consisting of 1024 data points (with a sample length of approximately 17 min) were seg-

mented in 16 partitions of 64 points yielding 32 degrees of freedom and frequency resolution of 0.015625 Hz. A fast Fourier transform (FFT) was applied to all three time series employing a Hanning window and 50 percent overlap between adjacent segments.

For convenience, the three time series η , η_x , and η_y are denoted 1, 2, and 3, respectively. The Fourier transform of the cross-correlation functions between them yields the cross-spectral density $S_{jk}(f)$:

$$S_{jk}(f) = C_{jk}(f) + iQ_{jk}(f)$$

where $j, k = 1, 2$, and 3 , $C_{jj}(f)$ is the auto-spectrum, $C_{jk}(f)$ is the co-spectrum, and $Q_{jk}(f)$ is the quad-spectrum ($Q_{jj}(f) = 0$).

Following the original work of Longuet-Higgins et al. [18], applying auto and cross-spectral analysis on the three components yields

$$\begin{aligned} C_{11}(f) &= \int_{-\pi}^{+\pi} S(f, \theta) d\theta, \\ C_{22}(f) &= k^2 \int_{-\pi}^{+\pi} S(f, \theta) \cos^2(\theta) d\theta, \\ C_{33}(f) &= k^2 \int_{-\pi}^{+\pi} S(f, \theta) \sin^2(\theta) d\theta, \\ Q_{12}(f) &= k \int_{-\pi}^{+\pi} S(f, \theta) \cos(\theta) d\theta, \\ Q_{13}(f) &= k \int_{-\pi}^{+\pi} S(f, \theta) \sin(\theta) d\theta, \\ C_{23}(f) &= k^2 \int_{-\pi}^{+\pi} S(f, \theta) \sin(\theta) \cos(\theta) d\theta, \\ C_{12}(f) &= C_{13}(f) = Q_{23}(f) = 0 \end{aligned} \quad (5)$$

where

$$\theta = \tan^{-1} \frac{Q_{13}(f)}{Q_{12}(f)} \quad (6)$$

is the direction of propagation, and where the wave number is represented by

$$k = k(f) = \left\{ \frac{C_{22}(f) + C_{33}(f)}{C_{11}(f)} \right\}^{1/2} \quad (7)$$

The directional wave spectrum $S(f, \theta)$ describes how energy propagates in frequency and direction, and is represented by the product of the one-dimensional spectrum $S(f)$ and the energy-spreading function $D(f, \theta)$ as

$$S(f, \theta) = S(f) D(f, \theta) \quad (8)$$

where

$$\int_{-\pi}^{+\pi} D(f, \theta) d\theta = 1 \quad (9)$$

The spreading function $D(f, \theta)$ can be expanded as a Fourier series as follows:

$$D(f, \theta) = \frac{1}{\pi} \left\{ \frac{1}{2} + \sum_{n=1}^2 (a_n(f) \cos n\theta + b_n(f) \sin n\theta) \right\} \quad (10)$$

where

Table 1 Occurrence of spectral peaks in Campos Basin

$Hs(m)$	1 peak	2 peaks	3 or more	num. obs.	% obs.
$Hs \leq 1.5$	236	533	706	1475	25.40
$1.5 < Hs \leq 2.5$	776	1468	1176	3420	58.89
$2.5 < Hs \leq 3.5$	312	298	121	731	12.59
$3.5 < Hs \leq 4.5$	116	24	3	143	2.46
$4.5 < Hs \leq 5.5$	25	9	0	34	0.59
$5.5 < Hs$	4	0	0	4	0.07
num. obs.	1469	2332	2006	5807	
% obs.	25.30	40.16	34.54		100

$$\begin{aligned} a_1(f) &= \int_{-\pi}^{+\pi} D(f, \theta) \cos(\theta) d\theta = \frac{Q_{12}(f)}{kC_{11}(f)}, \\ b_1(f) &= \int_{-\pi}^{+\pi} D(f, \theta) \sin(\theta) d\theta = \frac{Q_{13}(f)}{kC_{11}(f)}, \\ a_2(f) &= \int_{-\pi}^{+\pi} D(f, \theta) \cos(2\theta) d\theta = \frac{C_{22}(f) - C_{33}(f)}{k^2 C_{11}(f)}, \\ b_2(f) &= \int_{-\pi}^{+\pi} D(f, \theta) \sin(2\theta) d\theta = \frac{2C_{23}(f)}{k^2 C_{11}(f)}. \end{aligned} \quad (11)$$

The Fourier expansion (10) is truncated at $n=2$ given the limitation that only three components are measured by the buoy, i.e., the vertical displacement and two slopes. This truncation causes negative lobes in $D(f, \theta)$, which is always positive definite. Longuet-Higgins et al. [18] proposed the application of weighting functions to solve this limitation, which causes an undesirable broadening of the spreading function.

The application of Fourier analysis to obtain the spreading function $D(f, \theta)$ is limited to the small number of components used in the expansion. Due to this limitation, alternative approaches are applied to estimate the spreading function; among them, the nonparametric maximum entropy method (MEM) produces reasonable results, particularly in the reconstruction of directional double-peaked wave spectra (Lygre and Krogstad [19]). MEM is a nonparametric model in the sense that there is no fixed analytical form. In contrast to MEM, parametric models such as the forms proposed by Longuet-Higgins et al. [18] and Donelan et al. [2] have a predetermined form and a controlling parameter (respectively, s and β) which is dependent on a single-peak frequency. These parametric models are not consistent with reconstructing the two-dimensional spectrum $S(f, \theta)$ when windsea and swell co-exist, since they attempt to fit a single peak centered between the two wave directions (Young [20]). Hence, MEM is more suitable for the present purpose and was applied for the estimation of $D(f, \theta)$.

Violante-Carvalho [21] presented a comprehensive description of the wave climate in Campos Basin. In that work, some wave field evolution patterns are identified and associated with the meteorological situations which generated them. The decomposition of the spectra into different wave systems (windsea, swell, and quasi-sea) and the information on how energy is distributed in terms of frequency and direction is presented in statistical tables for different seasons of the year. Some of the characteristics of the wave spectra analyzed appear in Table 1, which shows the occurrence of spectral peaks as a function of the significant wave height Hs . The dataset spans a period of 26 mo, during which 5807 wave spectra were measured. The data indicate that about 25 percent of the spectra are unimodal, whereas the vast majority presents two or more peaks.

The frequent occurrence of multi-modal spectra can be explained on the grounds of the meteorological conditions encountered in the region. The semi-stationary South Atlantic high-pressure center has a fundamental role in the wave climate in Campos Basin. The anticyclonic circulation is associated in the

area with northerly or easterly winds generating windsea with peak periods spanning from 3 to 5 s and significant wave height from 0.25 to 1.25 m. Another wave system is also associated with the semi-stationary anticyclone, although it is not generated by the local wind. Due to the curved form of the isobars around the high-pressure center, a wave system is generated nearby by winds having directions in general 20 to 40 deg higher (clockwise) than the wind direction locally measured (winds slightly more easterly). In this scenario, a young swell or a *quasi-sea* with higher fetches and more energetic waves is generated with typical values of peak period ranging from 5 to 8 s and H_s from 0.5 to 2 m. Campos Basin is also strongly affected all the year round by swell from the south quadrant with typical peak periods ranging from 8 to 15 s and values of significant wave height from 0.5 to 2 m.

The Method for the Spectral Fitting

Formulations of the type (2, 3, 4) are for unimodal spectra and cannot represent the sea state when there is windsea and swell. However, they form the theoretical basis for the representation of multi-peaked spectra [22–24]. In these papers, it is assumed that the wave spectrum is composed of wave systems generated by different sources, and there is no interaction between each spectral component. Hence, the methods for representing a bimodal spectrum assume that the one-dimensional spectrum $S(f)$ is the sum of a high-frequency S_{hf} and a low-frequency S_{lf} component yielding

$$S(f) = S_{hf} + S_{lf}. \quad (12)$$

Ochi and Hubble [22] proposed a method for the adjustment of bimodal spectra where each part is expressed by three parameters. They represented the high and low spectral components using a modified version of the Pierson-Moskowitz spectrum. Guedes Soares [23] adopted the same approach to model double-peaked wave spectra, but using a spectrum derived from JONSWAP. The justification for modeling the high-frequency part of the spectrum is straightforward since the JONSWAP form was originally developed for windsea. However, this spectral formulation does not apply for swell. His explanation for adjusting the low frequency using a JONSWAP model was the narrowness of the swell, which can be well represented by the sharp spectral shape of (3).

The spectral representation suggested by Guedes Soares [23] seems to be more appropriate than that of Ochi and Hubble [22]. The good fit of the JONSWAP-type formulations (3 and 4) has been confirmed by numerous experiments and the power laws for wave growth were obtained by applying these spectral forms. Nevertheless, as has already been pointed out, the exponent n that best describes the high-frequency part of the spectrum in the form $S(f) \propto f^{-n}$ is not exactly known. McCarthy [24] also applied a modified version of JONSWAP to adjust double-peaked spectra. His approach was based on a more flexible spectral form where the logarithmic decay is allowed to vary, and he added a new parameter to adjust the low-frequency region before the peak.

In order to investigate the behavior of the high-frequency tail decay, we employed a spectral method based on JONSWAP using a variable exponent n

$$S(f) = (2\pi)\alpha g^2 (2\pi f)^{-n} \exp\left[-\frac{5}{4}\left(\frac{f}{f_p}\right)^{-4}\right] \gamma^{\exp[-(f-f_p)^2/2\sigma^2 f_p^2]}. \quad (13)$$

The values of σ are the same as in (3), and hence there are four parameters to be determined for the spectral adjustment: f_p , n , α , and γ .

The first step in the fitting procedure is to select the spectral peaks (see flow chart in Fig. 2). As discussed by Rodriguez and Guedes Soares [25], the correct identification of the points of maximum ordinates might be masked by random fluctuations arising from the spectral estimators. Pierson [26] pointed out that 70 percent of the spectral peaks that are four bands wide appear to present two peaks, but this is a consequence of sampling variability

and does not represent an underlying geophysical cause of double peaks. Therefore, this selection requires additional tests to identify whether a separate peak corresponds to a different wave system or not.

A balance must be found between resolution and variance, i.e., a compromise between increasing the number of degrees of freedom (and hence reducing the variance), and the capability to identify small nuances in the smoothed spectrum. The spectra with 32 degrees of freedom and 50 percent overlap yielded a satisfactory relation between resolution and variance, although additional criteria are applied to check whether the peaks are due to the sampling variability. Guedes Soares and Nolasco [27] presented some criteria for the selection of the peaks, of which two are based on the 90-percent confidence interval test (CIT). This first CIT imposes the condition that the two peaks will be accepted if the ordinate of lower limit of the 90 percent confidence interval of the greater peak is higher than the ordinate of the upper limit of the confidence interval of the trough between the peaks. In addition, they proposed a second CIT requiring that the ordinate of the trough between the two peaks must be lower than the lower limit of the 90-percent confidence interval of the smaller peak.

We adopted three criteria which must all be satisfied for the selection of the spectral peaks. The first criterion requires that the peaks must be 2-times the frequency resolution (0.03 Hz) apart. This was considered the best value after several tests using distances spanning from 0.02 to 0.05 Hz. The second criterion is based on the 90-percent confidence intervals and the peaks will be accepted if they satisfy the conditions of the first CIT in Guedes Soares and Nolasco [27]. The second CIT proposed by them is too restrictive, eliminating many peaks that were visually accepted. The third criterion applied for peak selection is that the ratio between the two spectral peaks must be less than 15. This requirement eliminates peaks that are below an arbitrary threshold value, and hence contaminated by background noise.

The next step in the fitting procedure is to adjust the high-frequency spectrum S_{hf} . The decay n is determined by logarithmic regression of the points greater than $2f_p$. To obtain α the value of γ was set to 1 in (13) because its effect at frequencies higher than $f/f_p > 1.37$ is negligible. The value of α was obtained by iterating from 0.0001 to 0.1 in order to minimize the total square error for the range between 1.37 and 2 times the peak frequency (Hasselmann et al. [28]). Once f_p , n , and α are known, γ is the last parameter to define S_{hf} . It was chosen as the value that produces the least-square error between measured and fitted spectra.

The adjusted high-frequency spectrum S_{hf} is subtracted from the measured spectrum yielding S_{lf} (12). The same process is repeated to obtain the parameters n , α , and γ for the low-frequency spectrum. These procedures outline the fitting of bimodal spectra.

The wave climate in Campos Basin is complex and spectra with three or more peaks are common, about one-third as indicated by Table 1. The fitting procedures for double-peaked wave spectra selects the two points of highest spectral ordinate. Many times, however, the windsea spectra in Campos Basin were not one of the two more energetic spectra. In such a case, the fitting procedure was extended to model a third peak, but only if the two selected ones do not correspond to the windsea (as defined in the next section). In addition, the third peak must satisfy the three criteria previously outlined for the spectral selection. A trimodal spectrum is given as

$$S(f) = S_{sea} + S_{hf} + S_{lf} \quad (14)$$

where in this situation the windsea is represented by S_{sea} , the *quasi-sea* by S_{hf} , and swell by S_{lf} . In a bimodal spectrum (12), in general, the windsea will be S_{hf} and the *quasi-sea* or swell will be represented by S_{lf} . Therefore, the wave spectrum generated by the local wind is always adjusted.

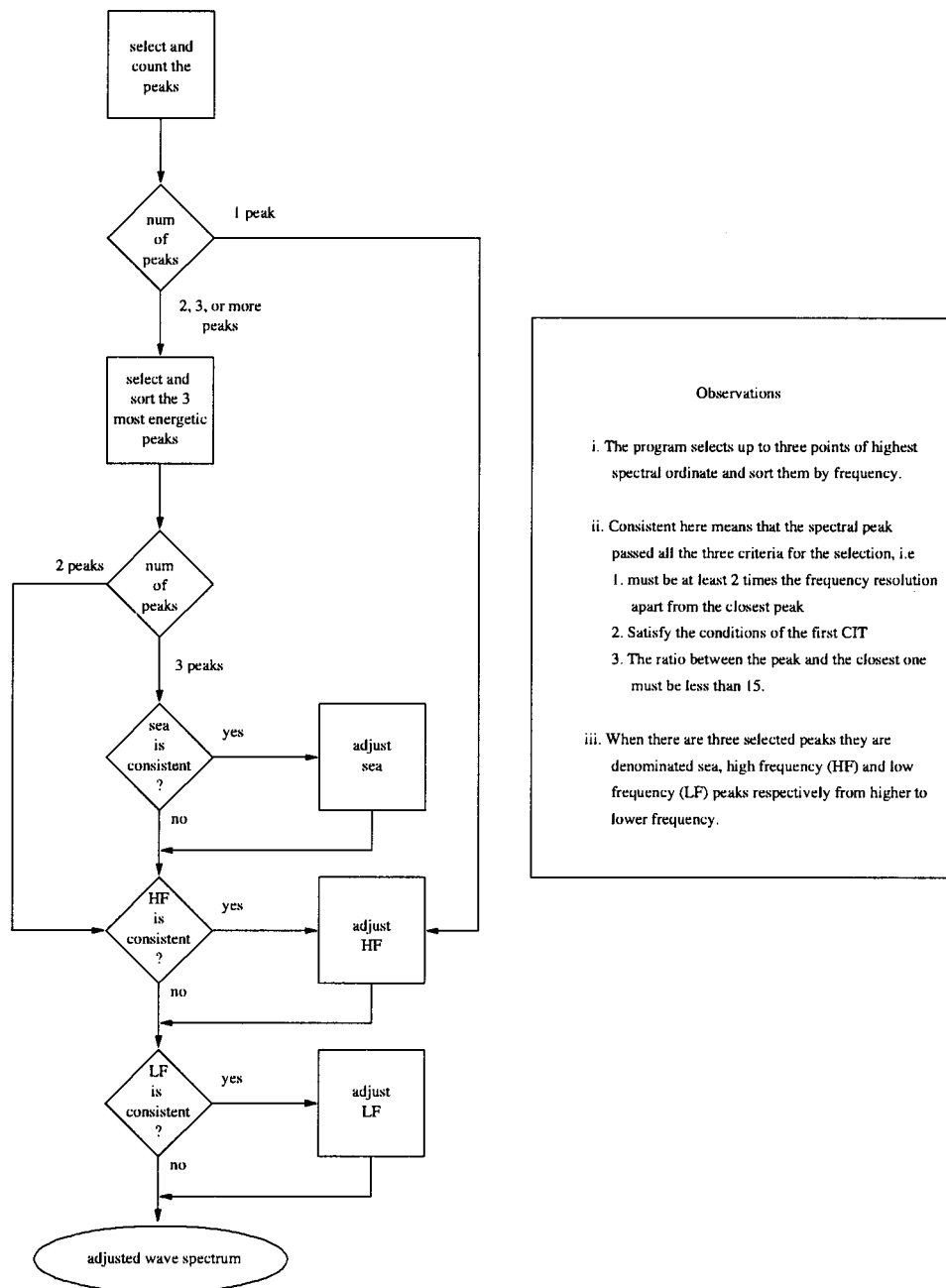


Fig. 2 Flow chart of the algorithm for the spectral adjustment of the 1-D wave spectrum

Examples of bimodal and trimodal adjustments of the 1-D spectrum are presented at the top panels in Fig. 3. On the left, the bimodal spectrum has a dominant high frequency peak from ENE at 0.1514 Hz (6.6 s) and an H_s value of 1.83 m, while the least energetic low-frequency peak (0.0859 Hz or 11.6 s) from the south quadrant has $H_s = 0.99$ m. The more complicated trimodal spectrum on the right side presents windsea, high and low-frequency peaks of 0.2119, 0.1318, and 0.0840 Hz (4.7, 7.6, and 11.9 s) and H_s values of 0.83, 0.54, and 0.69 m, respectively. Although the directional spectrum (bottom panels in Fig. 3) is not fitted, it is needed to identify which of the 1-D spectral peaks is in the direction corresponding to the windsea. The 2-D spectrum is also necessary to characterize the direction of the fitted wave systems, which is of great interest in terms of offshore operations in Campos Basin.

Windsea Growth

The knowledge of the evolution of wind generated waves is based on experiments carried out at limited fetches and small depths encountered in lakes and semi-closed basins. The influence of swell on wind wave growth in open ocean regions is more difficult to demonstrate, partially because of the necessity to isolate the windsea from the rest of the spectrum. In order to investigate the wave growth in a swell-dominated open ocean region, we examined a 26-mo series of wind and directional wave observations. Our considerations were limited to the cases of waves generated by steady wind conditions, and all the periods during which the wind direction varied by more than ± 20 deg from the mean direction were discarded. It is worth noting that swell was present in the data most of the time. Furthermore, we picked up

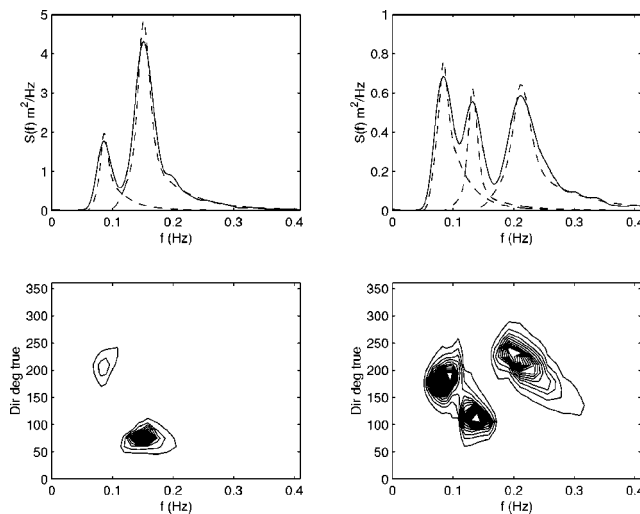


Fig. 3 Examples of the adjusted wave spectrum. Top panels show the measured 1-D spectrum (solid line) and the adjusted spectrum (dashed line) for a bimodal (left) and trimodal (right) case. Bottom panels show the respective directional or 2-D spectra calculated using the maximum entropy method.

only the periods in which the wind direction remained stable for at least five days to ensure that the wave field attained an equilibrium stage. The equilibrium of the wind with the steady wave field was fundamental to the analysis since the variation of the wind would contribute to increase the scatter of the data and complicate the interpretation.

The response of the heave-pitch-roll buoy to the sea movements is critical to the selection of the windsea, specially in the high-frequency region due to the presence of noise generated, for example, by currents and the mooring system (Tucker [29]). The size of the instrument and the accuracy of the procedure to evaluate the spectral estimators allowed the selection of the spectral peaks within the band from 0.04 to 0.35 Hz, whereas outside this range the instrumental response is questionable. This high-frequency limit is important to the selection of the windsea peak. To ensure that the third energetic peak is in fact windsea, further tests are applied, i.e., requiring that the direction of propagation of waves remains within a wind range and that α is above a windsea threshold level.

We concern ourselves with the growth of deepwater waves, and we therefore need reliable criteria to classify the windsea. The peaks that remain within a ± 30 deg window of the wind direction measured by the buoy are selected by the partitioning scheme. The quasi-sea spectrum may, however, occupy the wind-forced high-frequency region and may have wave directions close to the local wind. Hence, an additional criterion is necessary to provide isolation of the spectral peaks that lie in the region of windsea generation. The equilibrium range theory is employed to select the peaks whose high-frequency level α is higher than a threshold value. Examination of the data indicates that the wave components above $\alpha = 0.001$ are correlated to the wind energy transfer region, and so were classified as windsea.

The total number of selected windsea spectra was 243, and these were retained for further analysis. The most appropriate high-frequency decay was investigated applying a spectral form with a variable exponent n (13). Figure 4 shows the high-frequency decay for the cases classified as windsea. As can be seen, n exhibits a high variability, although most of the values lie between -4 and -6 . There is no clearly predominant decay that supports either the f^{-4} or f^{-5} form normally encountered in the literature as the best representation for the data analyzed. A unique value for n might not be appropriate and a spectral model with a variable high-frequency exponent such as (13) might be a better

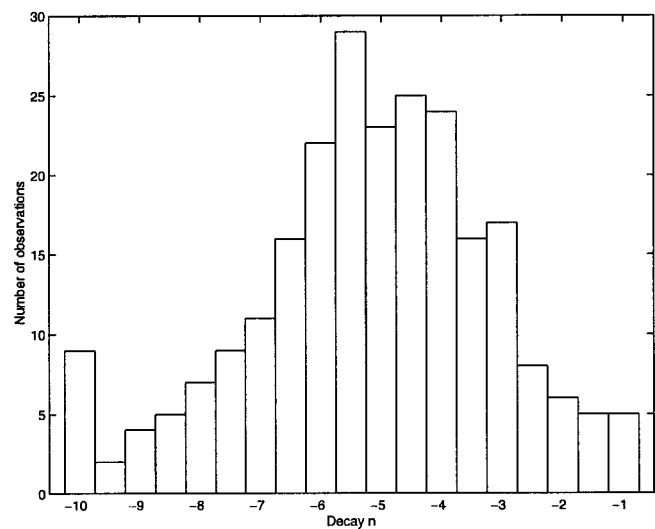


Fig. 4 Distribution of the exponent n for the 243 selected cases of windsea decay in the form $S(f) \propto f^{-n}$. All the cases with decays smaller than -10 were put together at the last bin.

option. We have found no clear evidence of dependence on other spectral parameters, and since the mean value is $n = -5.2$, a JONSWAP spectral form represented by (3) was applied to the data. Applying a constant decay allowed us to compare the results obtained through the analysis with previous works.

Experiments carried out in wave tanks are not conclusive if interactions between windsea and swell may affect the growth of waves. Some research indicates that windsea is intensified in the presence of opposed swell, whereas numerous experiments show an attenuation of the windsea energy (see a review of some wind-wave growth studies in Hanson and Phillips [30]). For practical purposes, it is fundamental to collect evidence that fetch laws which correlate, for instance wave energy and wind speed, may be applied in more realistic situations where swell is present.

The 243 values of the windsea high-frequency level α isolated by the method of spectral adjustment are plotted in Fig. 5 against

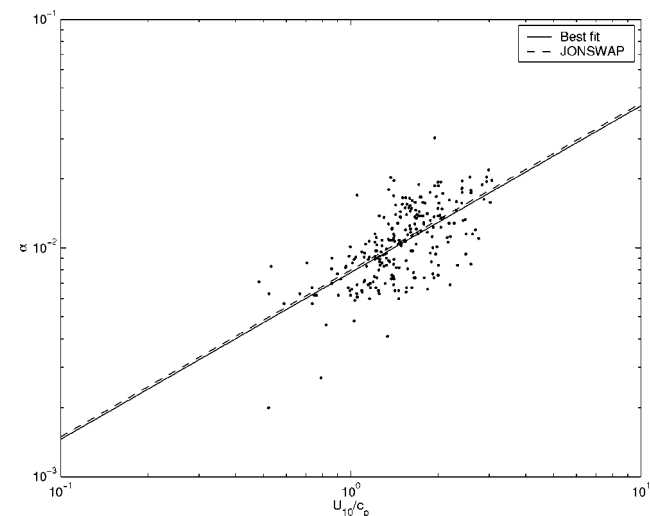


Fig. 5 Values of the high-frequency level α as a function of the reciprocal wave age U_{10}/c_p for a decay in the form $S(f) \propto f^{-5}$. The solid line is the best fit to the data $\alpha = 0.0078(U_{10}/c_p)^{0.7295}$ and the dashed line is the JONSWAP relationship for α in terms of U_{10}/c_p , $\alpha = 0.0080(U_{10}/c_p)^{0.73}$.

the reciprocal wave age U_{10}/c_p where U_{10} is the wind speed at 10 m height. A least-square regression of the data points is given by

$$\alpha = 0.0078 \left(\frac{U_{10}}{c_p} \right)^{0.7295} \quad (15)$$

with the correlation coefficient $r = 0.62$. The JONSWAP (Hasselmann et al. [7]) relationship for the variation of α with U_{10}/c_p is

$$\alpha_j = 0.0080 \left(\frac{U_{10}}{c_p} \right)^{0.73} \quad (16)$$

Despite the scatter in the data—as in similar investigations of spectral evolution—the fit is in very good agreement with the JONSWAP expression obtained in a fetch-limited swell-free environment. The parameter α depends on the wind speed and can be an indication of the stage of wave growth due to wind input. As a result of the dynamical processes of gain of energy from the wind, nonlinear interactions and energy lost due to white-capping the spectral peak moves to lower frequencies in a process of shape stabilization. The value of the phase velocity at the peak frequency c_p increases with fetch, and hence α decreases. The evidence that α has a geophysical meaning due to its response to the wind input during the process of wave growth and can be determined in an open-sea region dominated by swell can be useful in terms of wave forecasting. Once the wind vector is known, for instance by the measurement of a scatterometer onboard a satellite, the information of the corrected growth stage could be used to update ocean wave models.

Donelan et al. [2] proposed that $U_{10}/c_p = 0.83$ corresponds to the value of the spectrum at full development, where the spectral components below this value are classified as swell and above as windsea. Hence, the great majority of spectra in Fig. 5 are windsea, although 13 cases that were selected as windsea could be considered swell based on this separation criterion. In addition, many spectra are very mature—values of U_{10}/c_p ranging from 1 to 2—which is in accordance with the persistent steady wind conditions selected.

The slight differences between the idealized situations and our results may be related to the instrumental response and to the limitations in the partitioning method. Clearly, our measurements from an open region in the South Atlantic demonstrate that there is no significant effect of swell on the windsea growth.

Discussion

Although similar studies have been carried out previously, our investigation has allowed, for the first time, the study of the wave spectral evolution in a tropical oceanic region where swell is present. From an extensive database, we selected cases when the wave field attained an equilibrium stage with steady wind conditions. Based on Phillips' equilibrium range theory and applying a spectral method for the adjustment and partitioning, we were able to isolate the windsea from the swell-contaminated spectra in an offshore region in the South Atlantic. Our aim was to perform comparisons of the wave growth relationships obtained in idealized fetch-limited conditions in swell-free environments.

We have found no clear evidence of a unique value of n for representing the high-frequency exponent decay f^{-n} . In addition, no significant dependence of n on other spectral parameters was encountered. As most of the values lie between -4 and -6 , the spectral forms represented by either the JONSWAP or the Donelan models could be used. Actually, a spectral form which applies a constant exponent n might not be appropriate and a model such as (13) could be a better approach.

The best fit to the plot of the values of the parameter α against the reciprocal wave age U_{10}/c_p is in remarkable agreement with the JONSWAP relationship. The scatter in the data, which was expected to occur, is mainly due to the uncertainty of instrumental response in the high-frequency band of the spectrum close to the Nyquist frequency. The correlation obtained and the close agree-

ment with the JONSWAP power-law gave us confidence in the spectral technique for the adjustment and partitioning of the spectrum components. One must bear in mind the great differences among the experiments, i.e., unimodal spectra measured in a coastal area without swell and multi-modal spectra from a tropical oceanic region where swell is present and is responsible for the greatest amount of energy in the spectra.

Our results are in accordance with idealized fetch-limited experiments in sheltered areas. This fact gives strong evidence that the presence of swell has no significant influence on the wave growth and that power-laws proposed for unimodal spectra can be applied to practical situations in the open sea. Furthermore, the parameter α associated with the windsea part of the spectrum can be determined from multi-modal spectra. This indication might have practical applications in wave prediction models since α is correlated with the wind and is fetch dependent. The possibility to correlate the wind speed with the stage of windsea evolution could be used to update wave models improving the forecasting.

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